ASSIGNMENT - 03

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1) What is DNS (Domain Name System)

DNS (Domain Name System) is a system that converts human-readable domain names into IP addresses so computers can communicate over the internet.

Example :

when a user types www.google.com, DNS translates it into an IP address like 142.250.183.206.

- Without DNS, users would need to remember numerical IP addresses instead of simple domain names.
- DNS works like a phonebook of the internet. It connects the domain name to the server IP address where the website is hosted.

2) What are theTypes of DNS

1) Recursive DNS Resolver

This server receives the user's request and searches for the correct IP address by contacting other DNS servers.

Example: DNS servers provided by internet service providers.

2) Root DNS Server

Root DNS servers are the top level of the DNS hierarchy. They direct queries to the correct Top Level Domain (TLD) servers.

Example: It helps locate servers for .com, .org, .net.

3) TLD (Top Level Domain) Server

TLD servers manage domains such as **.com, .edu, .org, .gov**. They provide the address of the authoritative DNS server.

4)Authoritative DNS Server

This server contains the actual DNS records of the domain and provides the final IP address.

3) What Are The Services of DNS

1. Name Resolution

DNS converts domain names into IP addresses.

<u>Example:</u>

www.amazon.com → 176.32.103.205

2. Host Aliasing

DNS allows one domain name to have multiple aliases.

<u>Example:</u>

www.example.com and example.com may point to the same server.

3. Mail Server Identification

DNS helps locate the mail servers responsible for receiving emails for a domain.

Example:

gmail.com → mail servers for Gmail.

4. Load Distribution

DNS can distribute traffic across **multiple servers** to improve performance and availability.

Example :

Large websites like Google use multiple servers worldwide.

4)What Are The Records of DNS

- **A Record (Address Record)**

Maps a domain name to an IPv4 address.

<u>**Example:**</u>

example.com → 192.168.1.1

- **AAAA Record**

Maps a domain name to an IPv6 address.

<u>**Example:**</u>

example.com → 2001:db8::ff00:42:8329

- **CNAME Record (Canonical Name)**

Used to create an alias for another domain name.

Example:

www.example.com → example.com

- **MX Record (Mail Exchange)**

Specifies the mail server responsible for receiving emails for a domain.

Example:

gmail.com → mail.gmail.com

- **NS Record (Name Server)**

Specifies which DNS server is authoritative for a domain.

Example:

ns1.example.com

- **TXT Record**

Stores text information, often used for security verification like SPF and DKIM.

Example:

Used to verify domain ownership in services like Google.

THANK

YOU